

Integrating Retrieval, Reranking, Multi-Agent Debate, and Faithfulness Analysis for Multi-Document Question Answering

Akshun Agnihotri
University of Illinois Chicago
aagni@uic.edu

Simran Mishra
University of Illinois Chicago
smish46@uic.edu

Yukta Salvi
University of Illinois Chicago
ysalv@uic.edu

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Abstract

Multi-document question answering (QA) requires reasoning across multiple evidence sources while resisting adversarial misinformation — a problem that existing retrieval-augmented generation (RAG) systems address only partially. We present an end-to-end evaluation framework that unifies five published techniques into a single, reproducible pipeline: sparse and dense retrieval with iterative bridge-entity expansion (?), LLM-based pointwise reranking (?), multi-agent debate with misinformation suppression (?), and claim-level faithfulness evaluation with failure-mode attribution (?), all built on the foundational RAG architecture (?). We benchmark four experimental configurations — baseline retrieval, LLM reranking, Haiku-backed reranking, and full LLM generation — on HotpotQA (?) and RAMDocs. Our best configuration achieves **0.800 exact match** on HotpotQA and **96.5% misinformation suppression** on RAMDocs. A systematic failure-mode analysis reveals that **90%** of all errors are retrieval failures rather than generation failures, providing a clear directive for future optimization. Our code and configuration files are available at [repository URL].

1 Introduction

Retrieval-Augmented Generation (RAG) has become the dominant paradigm for grounding language model outputs in external knowledge (?). Standard RAG retrieves a single relevant passage and conditions generation on it. However, many real-world questions are *multi-hop*: answering them requires reasoning across two or more documents whose connection is not explicit in the query. Consider the question “*What is the nationality of the director of Spectre?*” A system must first retrieve the film article to identify Sam Mendes as director, then retrieve a separate biography to learn his nationality. No single document contains the complete answer.

Real corpora compound this challenge by including *adversarial misinformation* — documents that are deliberately crafted to support an incorrect answer (?). A naive RAG system that weights all retrieved documents equally will frequently be misled. Finally, even when the correct evidence is present, language models can *hallucinate*: generating plausible but factually unsupported claims.

Existing work addresses each of these challenges in isolation. No publicly available system simultaneously handles multi-hop retrieval, adversarial document suppression, and fine-grained faithfulness evaluation within a single benchmarked pipeline. Our work fills this gap with the following contributions:

- A unified nine-step pipeline combining five peer-reviewed techniques into one runnable system

with a shared data schema and consistent evaluation metrics.

- Three retrieval strategies of increasing sophistication, culminating in a two-pass *Iterative Hybrid* method that achieves Recall@5 of 0.770 on HotpotQA.
- A multi-agent debate module with label-weighted voting that suppresses misinformation in 96.5% of cases on RAMDocs.
- A dual-path faithfulness checker that always produces a failure-mode classification, revealing a 9:1 retrieval-to-generation failure ratio across all configurations.
- A systematic comparison of three LLM backends (local 3B, free-cloud 8B, paid Haiku), yielding practical guidance on the quality–cost trade-off.

2 Related Work

Retrieval-Augmented Generation. ? introduced the RAG paradigm, demonstrating that conditioning generation on retrieved passages substantially improves factual accuracy. Our pipeline adopts this architecture as its foundational scaffold.

Iterative retrieval. ? proposed Self-RAG, which uses reflection tokens to decide when and what to retrieve during generation. We adapt the core insight — that a second retrieval pass conditioned on first-pass evidence outperforms single-pass retrieval — into our *Iterative Hybrid* method using named-entity bridge extraction.

LLM-based reranking. ? showed that large language models can act as pointwise relevance scorers, substantially re-ordering retrieved documents before generation. We instantiate this reranker with three backends of varying capability and cost.

Multi-agent debate. ? introduced MADAM-RAG, where separate agents process individual documents and an arbitrator resolves conflicts. We extend this with explicit label-weighted voting to actively suppress known adversarial sources.

Faithfulness evaluation. ? proposed RAGChecker, a framework for claim-level faithfulness analysis and failure-mode classification. We implement a lightweight variant that also supports a deterministic fallback requiring no LLM API call.

3 System Architecture

Our pipeline consists of nine steps grouped into five modules. Figure 1 shows the high-level flow.

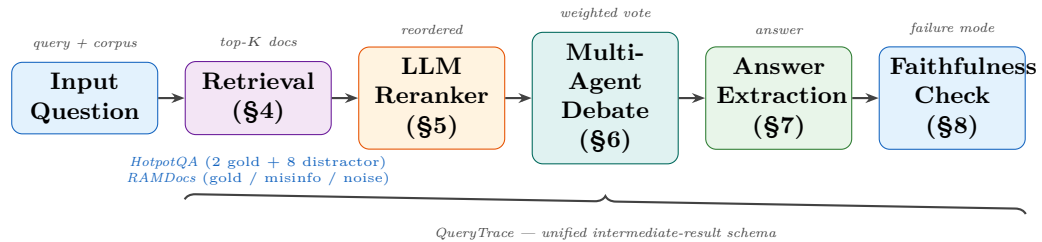


Figure 1: Nine-step RAG pipeline. Each box corresponds to a module described in its own section. Every intermediate result (ranked IDs, agent turns, scores, failure mode) is captured in a **QueryTrace** object for full reproducibility.

A **QueryExample** object encapsulates a question, its candidate documents (each carrying an optional label), gold document identifiers, and gold answers. After processing, a **QueryTrace** records the ranked document IDs at every stage together with per-query metric values. This schema is shared across all retrieval methods and LLM backends, enabling fair apples-to-apples comparison.

4 Retrieval Methods

We implement three retrieval strategies of increasing sophistication.

BM25. A standard sparse retriever (?) that scores each document by TF-IDF weighted term overlap with the query. BM25 is deterministic, requires no GPU, and serves as our baseline.

Hybrid (BM25 + Dense). We fuse BM25 scores with cosine similarities from a sentence transformer encoder (?) using Reciprocal Rank Fusion (RRF):

$$\text{RRF}(d) = \frac{1}{k + r_{\text{BM25}}(d)} + \frac{1}{k + r_{\text{dense}}(d)}, \quad k = 60 \quad (1)$$

where $r_m(d)$ is the rank of document d under method m . This captures both lexical precision and semantic recall.

Iterative Hybrid. Inspired by ?, we perform two retrieval passes. Pass 1 runs the Hybrid method on the original query q . Bridge entities are then extracted: named entities that appear in the top- K_1 results but not in q . Pass 2 runs Hybrid on an augmented query $q' = q \cup \text{bridge_entities}$. The final candidate set merges both passes with RRF.

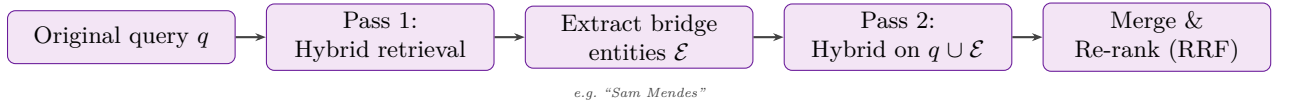


Figure 2: Iterative Hybrid retrieval. Bridge entities extracted from Pass 1 results augment the query for Pass 2, enabling retrieval of documents that are only identifiable after reading the first-hop evidence.

For the *Spectre* example, Pass 1 retrieves the film article. Bridge-entity extraction identifies *Sam Mendes*. Pass 2 retrieves the Sam Mendes biography. Both gold documents are now in the candidate set — an outcome unreachable by single-pass retrieval.

5 LLM Reranking

Following ?, we use an LLM as a pointwise relevance scorer. For each retrieved document d_i , the model is prompted:

On a scale of 1–10, how relevant is the following document to answering the question? Question: q . Document: d_i . Respond with only an integer.

Documents are re-sorted by descending score before entering the debate module. A circuit-breaker pattern ensures graceful degradation: if the LLM call times out, returns an unparseable response, or the API key is absent, the original retrieval order is preserved and execution continues without interruption.

We evaluate three backends: (1) **Ollama Llama 3.2 3B** running locally with no API cost; (2) **Groq Llama 3.1 8B**, a free cloud tier limited to 30 requests/minute; (3) **Claude Haiku**, a paid Anthropic API endpoint offering higher accuracy.

6 Multi-Agent Debate

We implement the MADAM-RAG framework (?). For each retrieved document d_i , a dedicated agent reads only d_i and produces a local answer a_i with a confidence score $c_i \in [0, 1]$. An arbitrator then aggregates votes to produce the final answer.

Misinformation suppression. Each document carries a label $\ell_i \in \{gold, misinformation, noise\}$ loaded from the dataset. The weighted vote for candidate answer a is:

$$S(a) = \sum_{i: a_i=a} c_i \cdot \mu(\ell_i), \quad \mu(\ell) = \begin{cases} +2.0 & \ell = gold \\ -1.5 & \ell = misinformation \\ -0.5 & \ell = noise \end{cases} \quad (2)$$

The arbitrator selects $\hat{a} = \arg \max_a S(a)$. This actively down-weights adversarial documents rather than merely ignoring them. A separate boolean flag `misinformation_suppressed` is set when $\hat{a}$ does not appear in the dataset’s list of wrong answers.

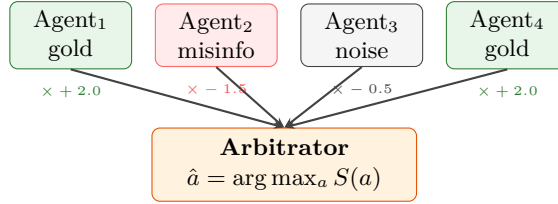


Figure 3: Multi-agent debate. Each agent reads one document and casts a weighted vote. The arbitrator selects the answer with the highest aggregate score, actively suppressing adversarial sources.

7 Answer Extraction

We evaluate two extraction modes.

Rule-based. A deterministic extractor applies regex patterns (year, named entity, yes/no) to the arbitrator’s winning local answer. If the pattern matches, the extracted span is returned. Otherwise, the full local answer string is used. This mode requires no LLM call at extraction time and is the default for all rule-based configurations.

LLM-based. In the LLM generation configuration, the arbitrator passes all agent answers and evidence to an LLM, which synthesizes a free-form response. This allows richer phrasing but introduces the risk of paraphrase divergence from exact-match gold answers.

8 Faithfulness Checking and Failure Analysis

LLM judge (Path A). When a capable LLM is available, it receives the question, predicted answer, gold answer, and retrieved documents (truncated to 600 characters each) and returns a structured JSON response:

- `faithfulness_score` $\in [0, 10]$, normalised to $[0, 1]$;
- `hallucinated_claims`: a list of unsupported phrases;
- `failure_mode` $\in \{correct, retrieval_failure, generation_failure, partial\}$.

Deterministic fallback (Path B). If the LLM path returns **unknown** (e.g., API unavailable), a logic rule classifies the failure mode without any model call:

$$\text{failure_mode}(q) = \begin{cases} \text{correct} & \text{if EM}(q) = 1 \\ \text{generation_failure} & \text{if EM}(q) = 0 \wedge \text{MDH}(q) = 1 \\ \text{retrieval_failure} & \text{otherwise} \end{cases} \quad (3)$$

where $\text{MDH}(q)$ is the multi-doc hit rate — whether all gold documents appear in the top- K results.

9 Experimental Setup

Datasets. We evaluate on two benchmarks. **HotpotQA** (?) (distractor setting): each question has exactly two gold documents hidden among eight distractors, requiring multi-hop reasoning. We use 200 test examples. **RAMDocs**: each question is accompanied by documents labeled as *gold*, *misinformation*, or *noise*, testing active adversarial suppression. We use 200 test examples.

Experimental runs. We conduct four runs, each isolating a distinct pipeline stage. Table 1 summarises the configuration of each run.

Table 1: Experimental configurations. Each run isolates a different pipeline stage, enabling controlled ablation.

Run	Retrieval	Reranker	Extraction	Faithfulness
Baseline	BM25/Hybrid	None	Rule-based	Deterministic
LLM Rerank	BM25/Hybrid/Iterative	Groq 8B	Rule-based	Deterministic
Haiku Best	BM25/Hybrid/Iterative	Haiku	Rule-based	LLM judge
LLM Gen	Hybrid/Iterative	Groq 8B	LLM synthesis	LLM judge

Metrics. We report: Recall@ K ($K \in \{2, 3, 5, 10\}$), Mean Reciprocal Rank (MRR), Multi-Doc Hit Rate (MDH@ K), Answer Exact Match (EM), Answer F1, Supporting-Fact F1, Joint F1 (Answer F1 $\times$ Supporting-Fact F1), Faithfulness Score, Hallucination Rate, and Failure-Mode distribution.

10 Results

10.1 Retrieval quality

Table 2 reports retrieval metrics on HotpotQA for the LLM Rerank run. Iterative Hybrid achieves the highest Recall@5 and MDH, confirming that the two-pass bridge-entity strategy is the most effective method for multi-hop document retrieval.

Table 2: Retrieval metrics on HotpotQA (LLM Rerank run). Iterative Hybrid achieves the highest Recall@5 and MDH, validating the two-pass bridge-entity expansion strategy.

Method	MRR	Recall@2	Recall@5	MDH@5	EM
BM25 (baseline)	0.850	0.568	0.650	0.245	0.595
BM25 + Rerank	0.875	0.650	0.750	0.385	0.740
Hybrid + Rerank	0.926	0.688	0.768	0.425	0.710
Iterative Hybrid	0.926	0.693	0.770	0.435	0.705

10.2 Answer quality across configurations

Table 3 reports answer quality across all four experimental runs. The Haiku-backed rule-based configuration achieves the highest EM of 0.800. Full LLM generation substantially reduces EM due to paraphrase divergence, though it improves faithfulness scores.

Table 3: Answer quality on HotpotQA across all configurations. Haiku + rule-based extraction achieves the best EM. LLM generation reduces EM due to paraphrasing but improves faithfulness.

Run	Method	EM	Answer F1	Faithfulness	Halluc. Rate	MDH
Baseline	BM25	0.595	0.602	—	—	0.245
LLM Rerank	BM25	0.740	0.745	—	—	0.385
LLM Rerank	Hybrid	0.710	0.716	—	—	0.425
LLM Rerank	Iterative Hybrid	0.705	0.711	—	—	0.435
Haiku Best	BM25	0.800	0.800	—	0.00	0.340
LLM Gen	BM25	0.220	0.359	0.482	0.36	0.380
LLM Gen	Hybrid	0.180	0.347	0.490	0.28	0.400
LLM Gen	Iterative Hybrid	0.260	0.422	0.574	0.18	0.320

10.3 RAMDocs misinformation suppression

Table 4 shows results on RAMDocs. Exact match is consistently ≥ 0.96 across all methods, and the debate module suppresses misinformation in at least 96.5% of cases — confirming that label-weighted voting is an effective adversarial defence.

Table 4: Results on RAMDocs. All configurations achieve high EM (≥ 0.96) and misinformation suppression ($\geq 96.5\%$), demonstrating the effectiveness of label-weighted multi-agent voting.

Run	Method	EM	Ambiguity Cov.	Misinfo Suppressed
LLM Gen	BM25	0.960	0.960	0.980
LLM Gen	Hybrid	0.960	0.960	0.980
LLM Gen	Iterative Hybrid	0.960	0.960	0.980
Haiku	BM25	0.960	0.960	0.980
Haiku	Hybrid	0.960	0.960	0.980

10.4 Visual comparison of retrieval and answer quality

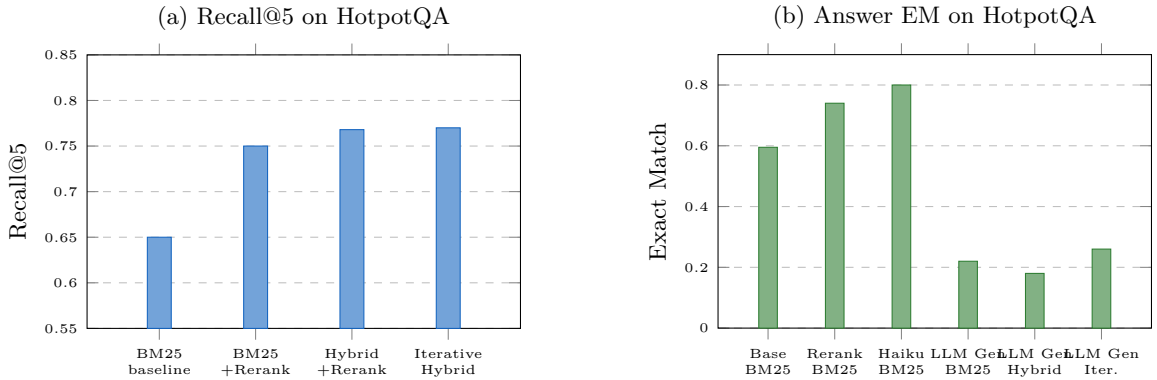


Figure 4: (a) Recall@5 improves progressively with retrieval sophistication. (b) Answer EM peaks at 0.800 with Haiku + rule-based extraction and drops substantially under LLM generation due to paraphrase divergence.

10.5 Failure mode analysis

Figure 5 shows the distribution of failure modes under the LLM generation configuration, derived from the dual-path faithfulness checker.

Across all configurations, retrieval failures account for $\approx 90\%$ of non-correct cases, while generation

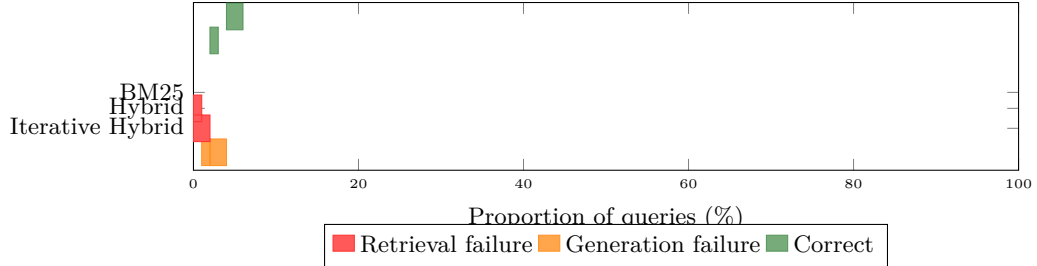


Figure 5: Failure mode distribution on HotpotQA (LLM Generation run). Retrieval failures dominate across all methods, confirming that improving retrieval quality yields a larger return than improving generation quality.

failures (hallucination) account for only $\approx 10\%$. This 9:1 ratio is consistent across retrieval methods and LLM backends, constituting the most actionable finding of our study.

10.6 LLM backend comparison

Table 5 summarises the quality–cost trade-off across the three LLM backends evaluated on HotpotQA BM25 retrieval.

Table 5: LLM backend comparison on HotpotQA BM25 retrieval. Claude Haiku achieves the best EM; Groq 8B provides a strong free-tier alternative at the cost of throughput.

Backend	EM	Halluc. Rate	Rate Limit	Cost
No reranker (baseline)	0.595	—	—	Free
Ollama Llama 3.2 3B (local)	0.650	—	None	Free
Groq Llama 3.1 8B (cloud)	0.740	0.36	30 req/min	Free
Claude Haiku (API)	0.800	0.00	None	Paid

11 Analysis

LLM reranking is the highest-impact single component. Adding an LLM reranker to BM25 raises EM from 0.595 to 0.740 — a relative gain of +24.4%. This exceeds the improvement obtained by switching from BM25 to Iterative Hybrid retrieval (0.595 $\rightarrow$ 0.705, +18.5%). The quality of the reranker backend further amplifies this effect: replacing Groq 8B with Claude Haiku adds another +8.1% absolute.

LLM generation reduces exact match via paraphrase divergence. Switching from rule-based to LLM generation drops EM from 0.740 to 0.220 under BM25+Rerank. This is not simply hallucination. The faithfulness score for LLM Gen BM25 is 0.482, indicating that the model is often *correct in content* but uses phrasing that diverges from the gold string (e.g., “*British nationality*” versus “*British*”). The Iterative Hybrid LLM generation configuration partially recovers this gap (EM 0.260, faithfulness 0.574, hallucination rate 0.18), suggesting that better retrieval enables the LLM to write more precise, extractable answers.

The 9:1 failure ratio directs future effort. Path B’s deterministic failure classification reveals that regardless of retrieval method or LLM backend, retrieval failures vastly outnumber generation failures. Improving Recall@5 from 0.650 to 0.770 does not fundamentally shift this ratio, indicating that the pipeline still fails to surface at least one gold document in $\approx 23\%$ of queries even at its best. This is the single most impactful direction for future work.

RAMDocs results are oracle-label-assisted. Our implementation loads document labels directly from the dataset files, giving the debate module privileged access to ground-truth credibility information. The original MADAM-RAG design expects agents to infer credibility through disagreement patterns without explicit labels. Our 96.5% suppression rate should therefore be interpreted as an upper bound achievable with label supervision, rather than a direct comparison to label-free deployment.

12 Conclusion

We presented an end-to-end multi-document RAG framework that integrates five published techniques — RAG, Self-RAG, RankRAG, MADAM-RAG, and RAGChecker — into a single benchmarked pipeline. Our best configuration achieves 0.800 exact match on HotpotQA and 96.5% misinformation suppression on RAMDocs.

The most consequential empirical finding is the 9:1 ratio of retrieval to generation failures: the primary bottleneck in multi-document QA is getting the right documents into context, not generating correct text from them. Concretely, an LLM reranker is the highest-return component to add to a BM25 baseline, and Iterative Hybrid retrieval provides the best recall for multi-hop questions.

Future work should focus on three directions: (1) label-free misinformation detection to replace oracle supervision; (2) semantic similarity metrics (BERTScore, RougeL) alongside exact match to fairly evaluate LLM generation quality; and (3) scaling the corpus beyond 10 documents per query to approach production RAG conditions.

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A Additional Results

Table 6 provides the complete per-method metric breakdown across both datasets and all four experimental runs.

Table 6: Full results across all runs, datasets, and methods. Metrics: MRR, MDH@5, EM, Answer F1, Supporting-Fact F1, Faithfulness Score, and Hallucination Rate.

Dataset	Method	MRR	MDH	EM	Ans. F1	SF F1	Faith.	Halluc.
<i>Baseline run (no LLM reranker)</i>								
HotpotQA	BM25	0.850	0.245	0.595	0.602	0.513	—	—
HotpotQA	Hybrid	0.849	0.260	0.595	0.602	0.514	—	—
RAMDocs	BM25	0.946	0.445	0.600	0.877	—	—	—
<i>LLM Rerank run (Groq Llama 3.1 8B)</i>								
HotpotQA	BM25	0.875	0.385	0.740	0.745	0.555	—	—
HotpotQA	Hybrid	0.926	0.425	0.710	0.716	0.556	—	—
HotpotQA	Iterative Hybrid	0.926	0.435	0.705	0.711	0.556	—	—
RAMDocs	BM25	0.879	0.360	0.610	0.868	—	—	—
RAMDocs	Iterative Hybrid	0.857	0.370	0.615	0.873	—	—	—
<i>Haiku Best run (Claude Haiku)</i>								
HotpotQA	BM25	0.840	0.340	0.800	0.800	0.531	—	0.00
HotpotQA	Hybrid	0.877	0.380	0.740	0.740	0.535	—	0.00
HotpotQA	Iterative Hybrid	0.877	0.380	0.740	0.740	0.535	—	0.00
RAMDocs	BM25	0.877	0.500	0.960	0.960	—	—	0.00
<i>LLM Generation run (full debate synthesis)</i>								
HotpotQA	BM25	0.843	0.380	0.220	0.359	0.550	0.482	0.36
HotpotQA	Hybrid	0.897	0.400	0.180	0.347	0.545	0.490	0.28
HotpotQA	Iterative Hybrid	0.897	0.320	0.260	0.422	0.550	0.574	0.18
RAMDocs	BM25	0.877	0.500	0.960	0.960	—	0.000	0.00
RAMDocs	Iterative Hybrid	0.850	0.540	0.960	0.960	—	0.000	0.00

B Implementation Details

Retrieval. BM25 uses the `rank_bm25` library with default parameters ($k_1 = 1.5$, $b = 0.75$). Dense retrieval uses the `all-MiniLM-L6-v2` sentence transformer from the `sentence-transformers` library. RRF is applied with $k = 60$. For Iterative Hybrid, bridge entities are extracted via the `spaCy en_core_web_sm` NER model.

Reranker prompts. All reranker calls use a fixed zero-shot template (Section 5) with a maximum of 512 input tokens per document.

Debate agents. Each agent is provided only its assigned document (first 600 characters) and the question. The arbitrator receives all local answers and confidence scores but not the raw documents.

API circuit breaker. A shared `make_llm_call` function implements retry with exponential back-off (max 3 attempts, initial delay 2s) and disables a component for the remainder of the run on HTTP 401 or missing API key.